

Chapter 8

RLC Circuits

Parallel RLC Circuits

Natural Response

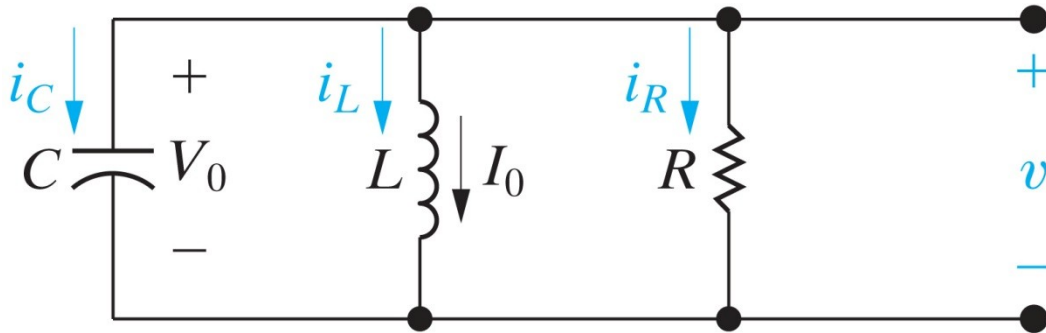


Figure: 08-01

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$V_0, I_0 \equiv$ Initial Voltage and Current

v is same for all elements

Write equation in terms of v , by Σi leaving top node

$$i_c + i_L + i_R = 0 \quad \left. \vphantom{i_c + i_L + i_R = 0} \right\} \text{KCL}$$

$$C \frac{dv}{dt} + \left[\frac{1}{L} \int_0^t v dt + I_0 \right] + \frac{v}{R} = 0$$

Substitute V/I
Relationships for
each element

How do we deal with the Integral?

Differentiate the
equation

Parallel RLC Circuits (Contd.)

$$C \frac{d^2 v}{dt^2} + \left[\frac{v}{L} + 0 \right] + \frac{1}{R} \frac{dv}{dt} = 0$$

Obtain a second order differential equation

$$\frac{d^2 v}{dt^2} + \left(\frac{1}{RC} \right) \frac{dv}{dt} + \frac{v}{LC} = 0$$

After simplification

2nd – order, ordinary differential equation with constant coefficients.

Solving 2nd – Order Circuits

1. Need a function whose 1st and 2nd – order derivatives are of the same form as the function.

2. 1st – Order circuits were exponential.

$$v = Ae^{st} \quad \text{where } A \text{ and } s \text{ are constants}$$

Assume:

- Assumed solution leads to the concept of the “Characteristic Equation”.

- Plug assumed solution into $\frac{d^2v}{dt^2} + \left(\frac{1}{RC}\right)\frac{dv}{dt} + \frac{v}{LC} = 0$

$$As^2e^{st} + \frac{As}{RC}e^{st} + \frac{A}{LC}e^{st} = 0 \quad \left. \vphantom{\frac{d^2v}{dt^2} + \left(\frac{1}{RC}\right)\frac{dv}{dt} + \frac{v}{LC} = 0} \right\} \text{After plugging in the solution}$$

$$Ae^{st} \left[s^2 + \left(\frac{1}{RC}\right)s + \frac{1}{LC} \right] = 0 \quad \left. \vphantom{Ae^{st} \left[s^2 + \left(\frac{1}{RC}\right)s + \frac{1}{LC} \right] = 0} \right\} \text{After simplification}$$

When is the above equation satisfied?

Solving 2nd – Order Circuits (Contd.)

When is

$$Ae^{st} \left[s^2 + \left(\frac{1}{RC} \right) s + \frac{1}{LC} \right] = 0$$

$$\left\{ \begin{array}{l} 1. \quad st \rightarrow -\infty \Rightarrow v(t) = 0 \\ 2. \quad A = 0 \Rightarrow v(t) = 0 \end{array} \right\} \text{ Trivial, not useful solutions}$$

$$3. \quad s^2 + \left(\frac{1}{RC} \right) s + \frac{1}{LC} = 0$$

Quadratic
Characteristic
Equation

General Procedure for 2nd – Order Circuits

$$a \frac{d^2 \chi}{dt^2} + b \frac{d\chi}{dt} + c\chi = 0 \quad \left\{ \begin{array}{l} a, b, c \text{ are constants} \end{array} \right.$$

$$as^2 + bs + c = 0 \quad \left\{ \begin{array}{l} \text{Characteristic Equation} \end{array} \right.$$

$$\chi(t) = Ae^{st} \quad \left\{ \begin{array}{l} \text{Solution} \end{array} \right.$$

Solving 2nd – Order Circuits (Contd.)

Quadratic Equation Review:

$$a\chi^2 + b\chi + c = 0 \quad \left. \vphantom{a\chi^2 + b\chi + c = 0} \right\} \text{a, b, c are constants}$$

$$\chi = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a} \quad \left. \vphantom{\chi = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}} \right\} \text{Roots of quadratic equation}$$

For the Characteristic Equation of the RLC Circuit

$$s_{1,2} = \frac{-\frac{1}{RC} \pm \sqrt{\left(\frac{1}{RC}\right)^2 - 4(1)\frac{1}{LC}}}{2(1)} \quad \left. \vphantom{s_{1,2} = \frac{-\frac{1}{RC} \pm \sqrt{\left(\frac{1}{RC}\right)^2 - 4(1)\frac{1}{LC}}}{2(1)}} \right\} \text{After using quadratic formula}$$

$$s_{1,2} = -\frac{1}{2RC} \left(\pm \sqrt{\left(\frac{1}{2RC}\right)^2 - \frac{1}{LC}} \right) \quad \left. \vphantom{s_{1,2} = -\frac{1}{2RC} \left(\pm \sqrt{\left(\frac{1}{2RC}\right)^2 - \frac{1}{LC}} \right)} \right\} \text{After simplification}$$

Solving 2nd – Order Circuits (Contd.)

Two Solutions for $s \Rightarrow$ Two solutions for $v(t)$ } $v(t) = Ae^{st}$

$$v_1(t) = A_1 e^{s_1 t} \quad \text{and} \quad v_2(t) = A_2 e^{s_2 t}$$

are both solutions

It can be shown that $v = v_1 + v_2$ is also a solution

$$v(t) = A_1 e^{s_1 t} + A_2 e^{s_2 t}$$

**General Solution
for Parallel RLC**

Initial Conditions



A_1 and A_2

Use initial conditions to
solve for A_1 and A_2
Need two initial conditions

Parallel RLC Solution for Voltage

Natural Response

$$1. \quad v(t) = A_1 e^{s_1 t} + A_2 e^{s_2 t} \quad \left. \vphantom{v(t)} \right\} \text{Solution}$$

$$2. \quad s_{1,2} = -\frac{1}{2RC} \pm \sqrt{\left(\frac{1}{2RC}\right)^2 - \frac{1}{LC}} \quad \left. \vphantom{s_{1,2}} \right\} \text{Characteristic roots}$$

$$s_{1,2} = -\alpha \pm \sqrt{\alpha^2 - \omega_0^2} \quad \left. \vphantom{s_{1,2}} \right\} \text{Complex Frequency}$$

$$3. \quad \alpha = \frac{1}{2RC} \quad \left. \vphantom{\alpha} \right\} \text{Neper Frequency}$$

$$4. \quad \omega_0 = \frac{1}{\sqrt{LC}} \quad \left. \vphantom{\omega_0} \right\} \text{Resonant Frequency}$$

Parallel RLC Solution for Voltage (Contd.)

$$\left. \begin{aligned} \frac{1}{RC} &\equiv \frac{1}{\Omega F} \equiv \frac{1}{\frac{V}{A} \cdot \frac{C}{V}} = \frac{A}{C} = \frac{C/s}{C} = \frac{1}{s} \end{aligned} \right\} \begin{array}{l} \text{Units for frequency} \\ \text{"1/seconds"} \end{array}$$

Neper Frequency

$$\left\{ \begin{array}{l} \alpha \text{ in units of } \frac{\text{rad}}{\text{sec}} \end{array} \right. \quad \text{rad are dimensionless}$$

$$f = \frac{\text{cycles}}{\text{sec}} \quad \text{cycles are dimensionless}$$

$$\omega = 2\pi f \frac{\text{rad}}{\text{s}} \equiv \text{Angular Frequency}$$

3 Possible Outcomes for Roots

$$1. \quad \omega_0^2 < \alpha^2 \Rightarrow s_{1,2} \text{ Real}$$

} Overdamped

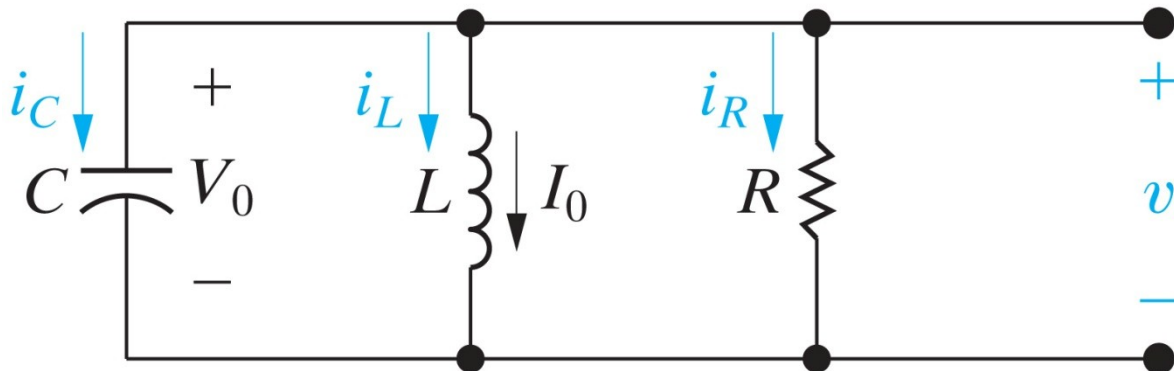
$$2. \quad \omega_0^2 > \alpha^2 \Rightarrow s_{1,2} \text{ Complex}$$

} Underdamped

$$3. \quad \omega_0^2 = \alpha^2 \Rightarrow s_{1,2} = -\alpha \text{ Repeated}$$

} Critically damped

Overdamped Voltage Response



$$\omega_0^2 < \alpha^2$$

Roots are real

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$$v(t) = A_1 e^{s_1 t} + A_2 e^{s_2 t} \quad s_1, s_2 \text{ are roots}$$

Initial Conditions

→ A_1 and A_2

$$v(0^+) \text{ and } \frac{dv(0^+)}{dt}$$

Need two initial conditions

$v(0^+)$ and $\frac{dv(0^+)}{dt}$
can be used to
find A_1 and A_2

$$v(0^+) = A_1 + A_2 \equiv V_0$$

Voltage across Capacitor

$$\frac{dv(0^+)}{dt} = s_1 A_1 + s_2 A_2 \equiv \frac{1}{C} i_C(0^+)$$

Current in the Capacitor

Overdamped Voltage Response (Contd.)

Find $i_c(0^+)$ $\longrightarrow$ $i_c(0^+) + i_L(0^+) + i_R(0^+) = 0$ } **KCL**

$$i_c(0^+) = -I_0 - \frac{V_0}{R}$$

Initial Capacitor Current in terms of Initial Inductor Current and Initial Voltage across Resistor

$$\frac{dv(0^+)}{dt} = \frac{1}{C} i_c(0^+) = \frac{1}{C} \left(-I_0 - \frac{V_0}{R} \right)$$

$\frac{dv(0^+)}{dt}$ in terms of I_0 and V_0

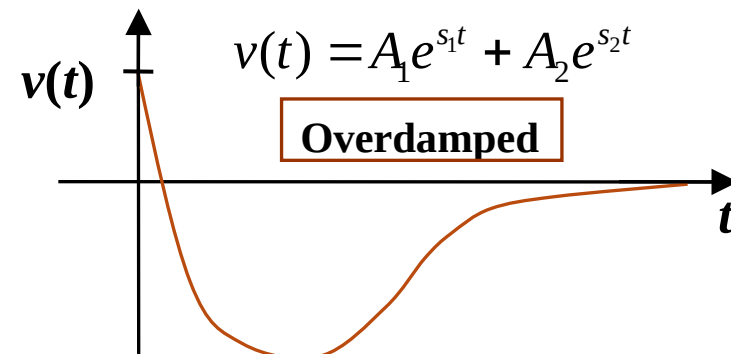
Solution Method

1. Find $s_1, s_2 \equiv \text{fcn}(R, L, \text{ and } C)$ } **Roots**

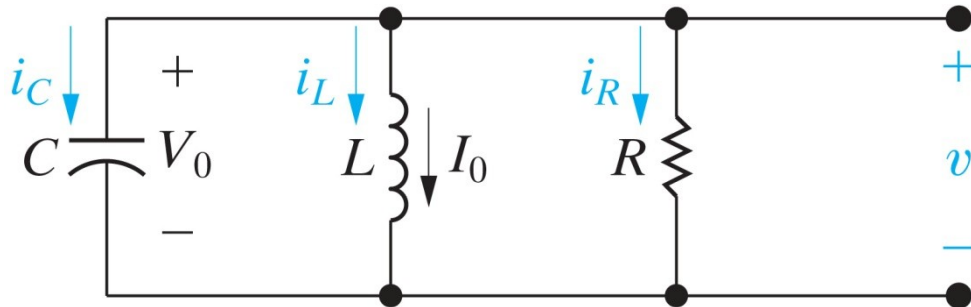
2. Find $v(0^+)$ and $dv(0^+)/dt$ } **Determine Initial Conditions**

3. Use $v(0^+)$ and $dv(0^+)/dt$ to find A_1, A_2

4. Plug s_1, s_2, A_1, A_2 into solution for $v(t)$



Drill Exercise: Find $v(t)$ for $t \geq 0$



$$R = 400 \, \Omega$$

$$I_0 = -4 \, \text{A}$$

$$L = 50 \, \text{mH}$$

$$V_0 = 0 \, \text{V}$$

$$C = 50 \, \text{nF}$$

Figure: 08-01

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$$\boxed{\omega_0^2 < \alpha^2} \left\{ \begin{aligned} \alpha &= \frac{1}{2RC} = \frac{1}{2(400)50 \times 10^{-9}} = 25,000 \, \text{rad/s} \\ \omega_0 &= \frac{1}{\sqrt{LC}} = \frac{1}{(50 \times 10^{-3})(50 \times 10^{-9})^{1/2}} = 20,000 \, \text{rad/s} \end{aligned} \right.$$

$\therefore$ Overdamped

$$s_{1,2} = -\alpha \pm \sqrt{\alpha^2 - \omega_0^2} \Rightarrow \left. \begin{aligned} s_1 &= -10 \, (\text{Krad/s}) \\ s_2 &= -40 \, (\text{Krad/s}) \end{aligned} \right\} \begin{array}{l} \text{Roots obtained} \\ \text{from quadratic} \\ \text{formula} \end{array}$$

$$\text{a) } \boxed{v(0^+) = V_0 = 0} \left\{ \begin{array}{l} \text{Given} \end{array} \right.$$

Drill Exercise (Contd.)

b). Find $\frac{dv(0^+)}{dt}$ by using $i_C(0^+)$

$$i_L(0^+) = I_0 = -4 \text{ A} \quad \text{Given}$$

$$i_C(0^+) = i_L(0^+) - i_R(0^+) \quad \text{KCL}$$

$$= -(-4\text{A}) - 0 \quad \text{V}_0 = 0 \Rightarrow i_R(0^+) = 0$$

$$i_C(0^+) = 4 \text{ (A)}$$

$$\frac{dv(0^+)}{dt} = \frac{1}{C} i_C(0^+) = \frac{4\text{A}}{50 \times 10^{-9} \text{ F}}$$

$$\frac{dv(0^+)}{dt} = 8 \times 10^7 \text{ V/s}$$

c). Initial Conditions $\Rightarrow A_1$ and A_2

$$v(0^+) = 0 = A_1 + A_2$$

$$\frac{dv(0^+)}{dt} = 8 \times 10^7 = s_1 A_1 + s_2 A_2 = -10,000 A_1 - 40,000 A_2$$

$$A_1 + A_2 = 0$$

$$(\div 10\text{K}) \Rightarrow A_1 + 4A_2 = -8000$$

$$\Rightarrow A_1 = \frac{8000}{3} \quad A_2 = -\frac{8000}{3}$$

d). $v(t) = A_1 e^{s_1 t} + A_2 e^{s_2 t}$

**Final
Solution**

$$v(t) = \frac{8000}{3} [e^{-10,000t} - e^{-40,000t}] \text{ (V)} \quad t \geq 0$$

Underdamped Voltage Response

$$\omega_0^2 > \alpha^2$$



Roots are complex

$$s_{1,2} = -\alpha \pm \sqrt{-(\omega_0^2 - \alpha^2)}$$

$$= -\alpha \pm \sqrt{-1} \cdot \sqrt{\omega_0^2 - \alpha^2} \quad \left. \vphantom{\begin{matrix} s_{1,2} \\ = -\alpha \pm \sqrt{-1} \cdot \sqrt{\omega_0^2 - \alpha^2} \end{matrix}} \right\} \begin{array}{l} \text{Factor out the} \\ \text{complex number} \end{array}$$

$$\text{Let } j = \sqrt{-1}$$

$$\begin{bmatrix} s_1 = -\alpha + j\omega_d \\ s_2 = -\alpha - j\omega_d \end{bmatrix}$$

$$\begin{bmatrix} \omega_d = \sqrt{\omega_0^2 - \alpha^2} \\ \text{Damped Radian Freq.} \end{bmatrix}$$

$$v(t) = A_1 e^{(-\alpha + j\omega_d)t} + A_2 e^{-(\alpha + j\omega_d)t}$$

**Underdamped
Solution**

Underdamped Voltage Response (Contd.)

Using Euler's Identity

$$\Rightarrow e^{\pm j\theta} = \cos \theta \pm j \sin \theta$$

Substitute in expression for $v(t)$

$$1 \quad v(t) = e^{-\alpha t} [(A_1 + A_2) \cos(\omega_d t) + j(A_1 - A_2) \sin(\omega_d t)] \quad \left. \vphantom{v(t)} \right\} \text{After simplifying the expression for } v(t)$$

$$2 \quad \text{Let } B_1 = A_1 + A_2 \text{ and } B_2 = j(A_1 - A_2)$$

$$3 \quad v(t) = e^{-\alpha t} [B_1 \cos(\omega_d t) + B_2 \sin(\omega_d t)] \quad \left. \vphantom{v(t)} \right\} \text{Real Number Solution}$$

$$\left. \begin{array}{l} B_1 = A_1 + A_2 \\ B_2 = j(A_1 - A_2) \end{array} \right\} \begin{array}{l} \text{Both are real numbers} \\ \text{because} \\ A_1 = A_2^* \text{ Complex Conjugate} \end{array}$$

Find the equations for the Initial Conditions

$$v(0^+) = V_0 = B_1 \quad \left. \vphantom{v(0^+)} \right\} \text{Use } 3$$

$$\frac{dv(0^+)}{dt} = \frac{1}{C} i_C(0^+) = -\alpha B_1 + \omega_d B_2 \quad \left. \vphantom{\frac{dv(0^+)}{dt}} \right\} \text{Use } 3$$

After some Algebra  See next slide

Find Constants A_1 and A_2

General Solution

$$v(t) = e^{-\alpha t} [(A_1 + A_2) \cos(\omega_d t) + j(A_1 - A_2) \sin(\omega_d t)]$$

Initial Conditions

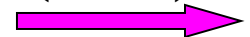
$$v(0^+) = V_0 = A_1 + A_2 \quad (1)$$

$$\begin{aligned} \frac{dv(t)}{dt} = & -\alpha e^{-\alpha t} (A_1 + A_2) \cos(\omega_d t) - e^{-\alpha t} \omega_d (A_1 + A_2) \sin(\omega_d t) \\ & - \alpha e^{-\alpha t} j(A_1 - A_2) \sin(\omega_d t) + e^{-\alpha t} j\omega_d (A_1 - A_2) \cos(\omega_d t) \end{aligned}$$

$$\frac{dv(0^+)}{dt} = \frac{1}{C} i_C(0^+) = -\alpha(A_1 + A_2) + j\omega_d(A_1 - A_2) \quad (2)$$

$$\text{Let } K = \frac{1}{C} i_C(0^+) \equiv \text{constant}$$

(Contd.)



Find Constants A_1 and A_2 (Contd.)

$$A_1 + A_2 = V_0 \quad (3)$$

$$- \alpha(A_1 + A_2) + j\omega_d(A_1 - A_2) = K \quad (4)$$

} Two equations for IC's

From (3) $A_2 = V_0 - A_1 \quad (5)$

$$A_1 - A_2 = A_1 - V_0 + A_1 = 2A_1 - V_0 \quad (6) \quad \text{Subtract (5) from } A_1$$

Insert (3) and (6) into (4)

$$- \alpha(V_0) + j\omega_d(2A_1 - V_0) = K \quad (7)$$

$$2A_1 - V_0 = \frac{K + \alpha V_0}{j\omega_d} = -j \left(\frac{K + \alpha V_0}{\omega_d} \right) \quad (8) \quad \text{Solve for } A_1$$

$$2A_1 = V_0 - j \frac{K + \alpha V_0}{\omega_d} \quad \text{Simplify}$$

$$A_1 = \frac{1}{2} \left(V_0 - j \frac{K + \alpha V_0}{\omega_d} \right) \quad (9)$$

Find Constants A_1 and A_2 (Contd.)

From (3) $A_2 = V_0 - A_1$ (10)

$$A_2 = V_0 - \frac{1}{2}V_0 + j \frac{K + \alpha V_0}{2\omega_d} \quad (11) \quad \left. \vphantom{A_2} \right\} \text{Substitute (9) into (10)}$$

$$A_2 = \frac{1}{2}V_0 + j \frac{K + \alpha V_0}{2\omega_d} \quad (12)$$

From the examination of (9) and (12), we see that:

$$A_1 = A_2^*$$

Since $A_1 = A_2^*$

Then $B_1 = A_1 + A_2 = V_0 \equiv \text{Real}$

and $B_2 = j(A_1 - A_2) = \frac{K + \alpha V_0}{\omega_d} \equiv \text{Real}$

Thus, the constants B_1 and B_2 are real.

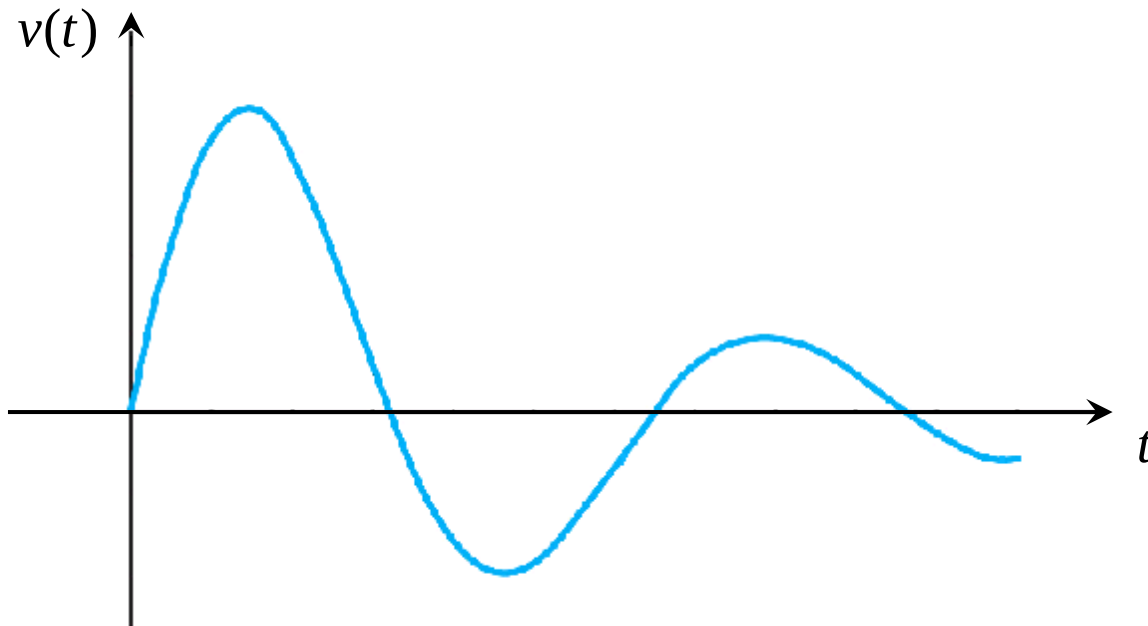
Underdamped Voltage Response (Contd.)

$$\omega_d = \sqrt{\omega_0^2 - \alpha^2}$$

$$\omega_0 = \frac{1}{\sqrt{LC}}$$

$$\alpha = \frac{1}{2RC}$$

Frequency parameter
related to 2nd – order
circuit parameters



Oscillatory Response

Typical
Underdamped
Response

Critically Damped Voltage Response

$$\omega_0^2 = \alpha^2 \quad \text{OR} \quad \omega_0 = \alpha$$

On the verge of oscillating

$$s_{1,2} = -\alpha$$

Advanced Solution

Proof beyond our scope

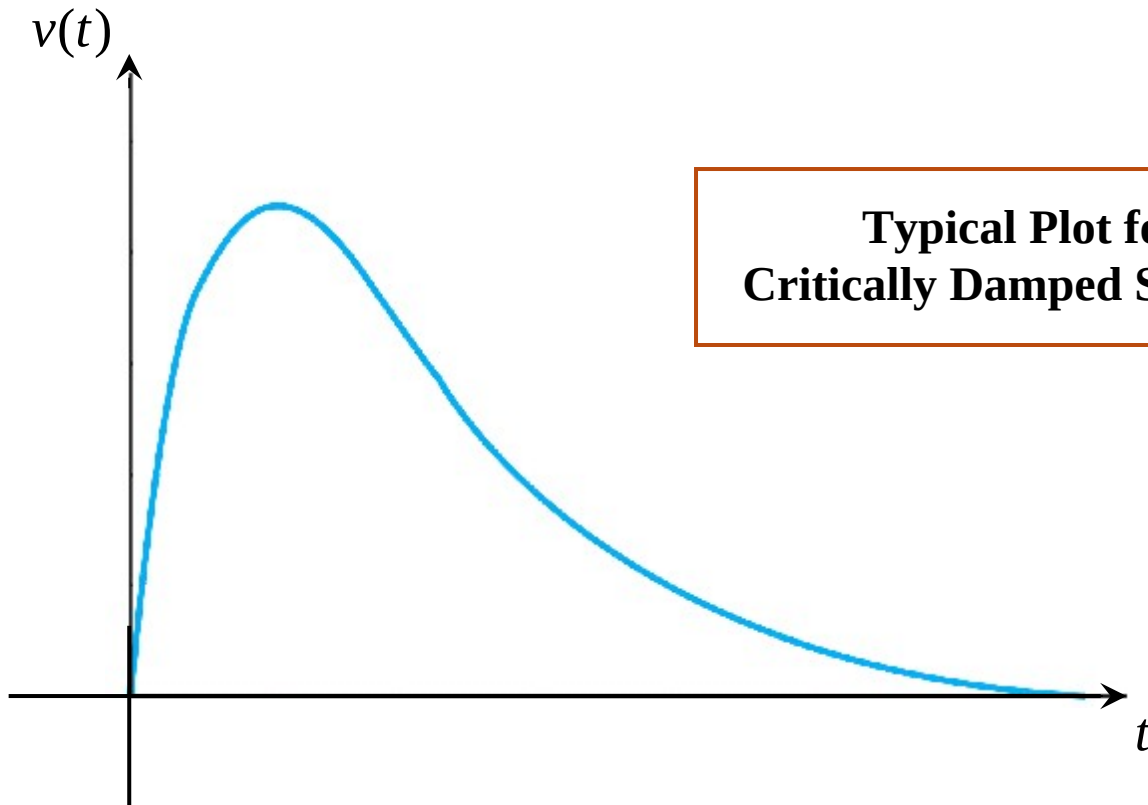
$$v(t) = D_1 t e^{-\alpha t} + D_2 e^{-\alpha t} \quad \left\{ \begin{array}{l} \text{Critically Damped Solution} \end{array} \right.$$

$$v(0^+) = V_0 = D_2$$

$$\frac{dv(0^+)}{dt} = \frac{i_C(0^+)}{C} = D_1 - \alpha D_2$$

Equations for
finding D_1 and D_2
from the Initial
Conditions

Critically Damped Voltage Response (contd.)



**Typical Plot for
Critically Damped Solution**

Summary of Parallel RLC Circuits

Natural Response

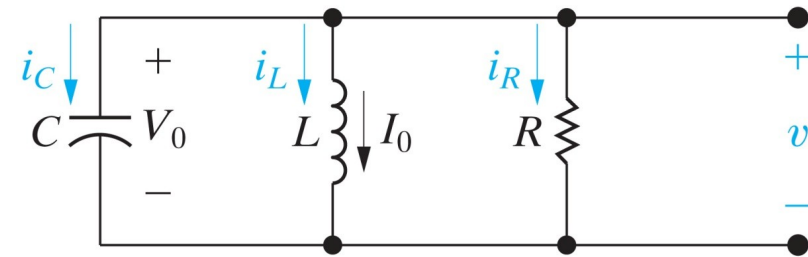


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$V_0, I_0 \equiv$ Initial Voltage and Current

v is same for all elements

$$v(0^+) = V_0$$

$$\alpha = \frac{1}{2RC} \quad \omega_0 = \frac{1}{\sqrt{LC}} \quad \frac{dv(0^+)}{dt} = \frac{1}{C} i_C(0^+) = \frac{1}{C} \left(-I_0 - \frac{V_0}{R} \right)$$

Overdamped $\omega_0^2 < \alpha^2$

$$s_1 = -\alpha + \sqrt{\alpha^2 - \omega_0^2}$$

$$s_2 = -\alpha - \sqrt{\alpha^2 - \omega_0^2}$$

$$A_1 = \frac{1}{s_1 - s_2} \left[\frac{dv(0^+)}{dt} - s_2 v(0^+) \right]$$

$$A_2 = \frac{1}{s_2 - s_1} \left[\frac{dv(0^+)}{dt} - s_1 v(0^+) \right]$$

$$V(t) = A_1 e^{s_1 t} + A_2 e^{s_2 t}$$

Underdamped $\omega_0^2 > \alpha^2$

$$\omega_d = \sqrt{\omega_0^2 - \alpha^2}$$

$$s_1 = -\alpha + j\omega_d$$

$$s_2 = -\alpha - j\omega_d$$

$$B_1 = v(0^+)$$

$$B_2 = \frac{1}{\omega_d} \left[\frac{dv(0^+)}{dt} + \alpha v(0^+) \right]$$

$$V(t) = e^{-\alpha t} \left[B_1 \cos(\omega_d t) + B_2 \sin(\omega_d t) \right]$$

Critically Damped $\omega_0^2 = \alpha^2$

$$\omega_0 = \alpha$$

$$s_1 = -\alpha$$

$$s_2 = -\alpha$$

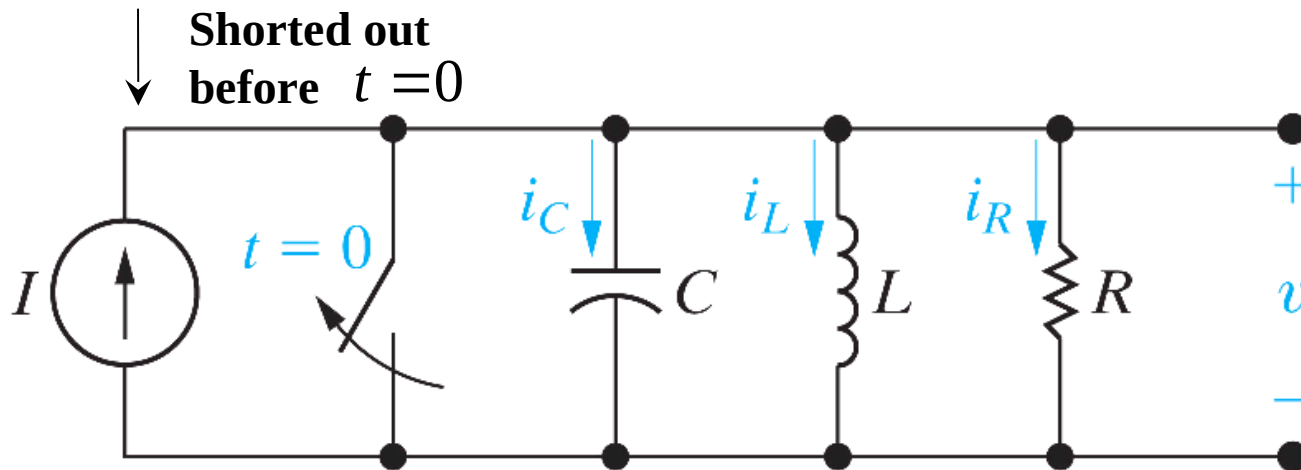
$$D_1 = \alpha D_2 + \frac{dv(0^+)}{dt}$$

$$D_2 = v(0^+)$$

$$V(t) = D_1 t e^{-\alpha t} + D_2 e^{-\alpha t}$$

Step Response of a Parallel RLC Circuit

Apply a DC current source
Find $i_L(t)$ for $t \geq 0$



$$\textcircled{1} \quad I = i_C + i_L + i_R \quad \left. \vphantom{I = i_C + i_L + i_R} \right\} \text{KCL}$$

$$\textcircled{2} \quad i_C = C \frac{dv}{dt} \quad i_R = \frac{v}{R} \quad \left. \vphantom{i_C = C \frac{dv}{dt} \quad i_R = \frac{v}{R}} \right\} \begin{array}{l} \text{Capacitor Law} \\ \text{Ohm's Law} \end{array}$$

$$\textcircled{3} \quad I = C \frac{dv}{dt} + i_L + \frac{v}{R} \quad \left. \vphantom{I = C \frac{dv}{dt} + i_L + \frac{v}{R}} \right\} \text{Substitute } \textcircled{2} \text{ into } \textcircled{1}$$

Step Response of a Parallel RLC Circuit (Contd.)

④ $v = L \frac{di_L}{dt} \Rightarrow \frac{dv}{dt} = L \frac{d^2 i_L}{dt^2}$ } Differentiate the Inductor Law

⑤ $I = LC \frac{d^2 i_L}{dt^2} + i_L + \frac{L}{R} \frac{di_L}{dt}$ } Substitute ④ into ③

Simplify ⑤ { $\frac{d^2 i_L}{dt^2} + \frac{1}{RC} \frac{di_L}{dt} + \frac{1}{LC} i_L = \frac{I}{LC}$ } Differential Equation for $i_L(t)$ “forced” by constant source I

Note the similarity to the natural response differential equation

$$\frac{d^2 v}{dt^2} + \left(\frac{1}{RC} \right) \frac{dv}{dt} + \left(\frac{1}{LC} \right) v = 0$$

Form is the same, except for constant.

Step Response of a Parallel RLC Circuit (Contd.)

In general, complete response of a 2nd – order system, with a constant forcing function is given by

Natural response + Forced response

$$i_L(t) = i_n(t) + I_f(t)$$

Diagram illustrating the components of the current response $i_L(t)$:

- Forced response** $I_f(t)$: Caused by a constant forcing function.
- Natural response** $i_n(t)$: Caused by initial conditions.

Coefficients in the analytical solutions are different from the case of zero forcing function.

Step Response of a Parallel RLC Circuit (Contd.)

Three cases: $\alpha = \frac{1}{2RC}$ $\omega_0 = \frac{1}{\sqrt{LC}}$

① Overdamped $\omega_0^2 < \alpha^2$

$$i_L(t) = A_1' e^{s_1 t} + A_2' e^{s_2 t} + \underline{\underline{I_f}}$$

Constant I_f Caused by a constant forcing function

$$s_{1,2} = -\alpha \pm \sqrt{\alpha^2 - \omega_0^2}$$

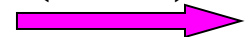
② Underdamped $\omega_0^2 > \alpha^2$

$$i_L(t) = e^{-\alpha t} [B_1' \cos(\omega_d t) + B_2' \sin(\omega_d t)] + \underline{\underline{I_f}}$$

$$\omega_d = \sqrt{\omega_0^2 - \alpha^2}$$

$$s_{1,2} = -\alpha \pm j\omega_d$$

(Contd.)



Step Response of a Parallel RLC Circuit (Contd.)

③ Critically damped $\omega_0^2 = \alpha^2$

$$i_L(t) = D_1' t e^{-\alpha t} + D_2' e^{-\alpha t} + \underline{\underline{I_f}}$$

Final condition:

Capacitor open
Inductor short



$$\left. \begin{aligned} V(t \rightarrow \infty) &= 0 \\ \therefore I_f &= i_L(t \rightarrow \infty) = I \end{aligned} \right\} \text{current source input}$$

Initial conditions: Used to determine scaling constants.

$$\begin{aligned} i_L(0^+) &= i_L(0^-) = I_0 \\ \frac{di_L(0^+)}{dt} &= \frac{V_0}{L} \end{aligned}$$



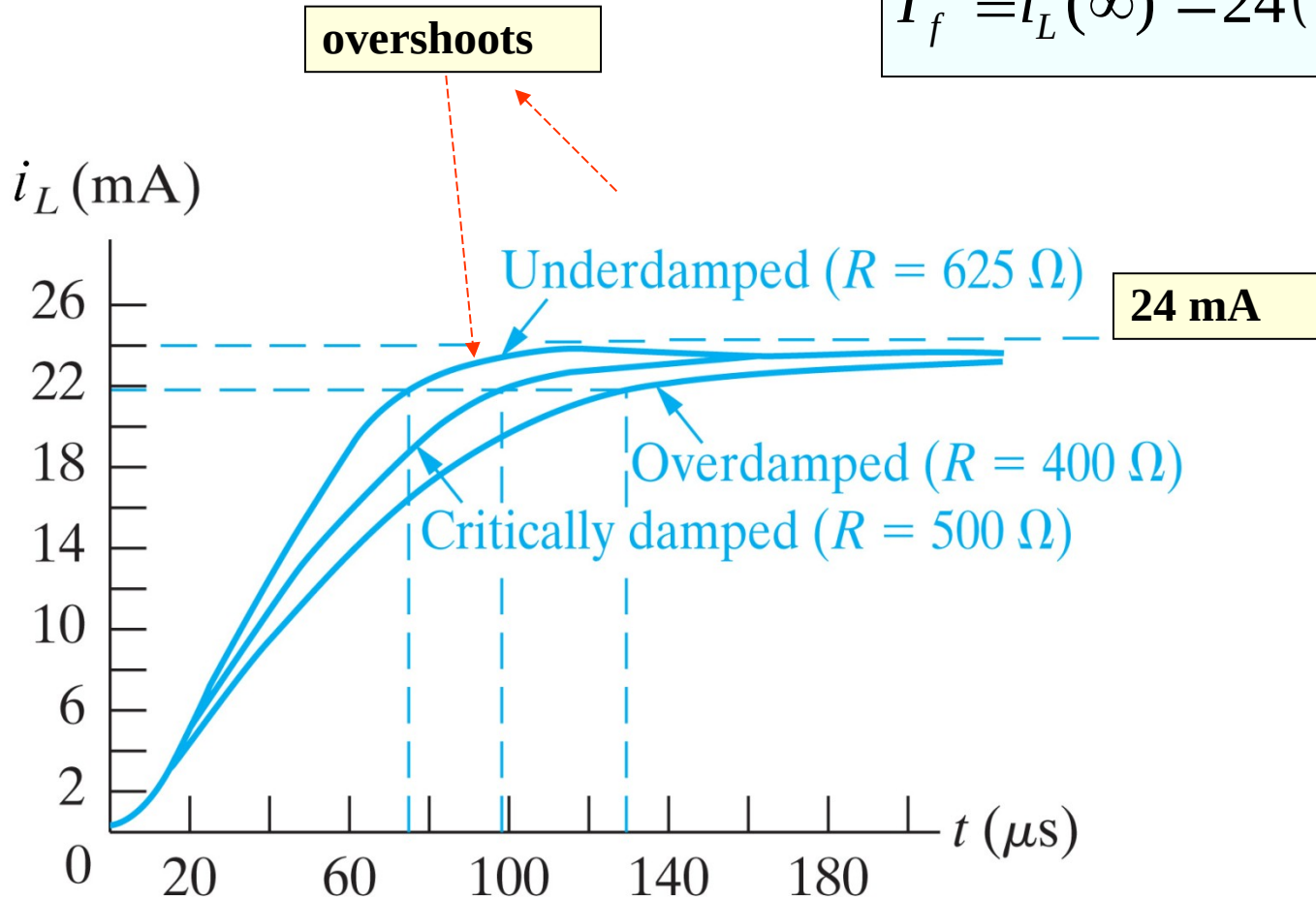
$$\text{since } v \equiv v_L = L \frac{di_L}{dt}$$

Note:

For initial currents, consider Inductor.
For initial voltage, consider Capacitor.

Step Response of Parallel RLC Circuit

$$I_f \equiv i_L(\infty) = 24 \text{ (mA)}$$



Study Suggestions:

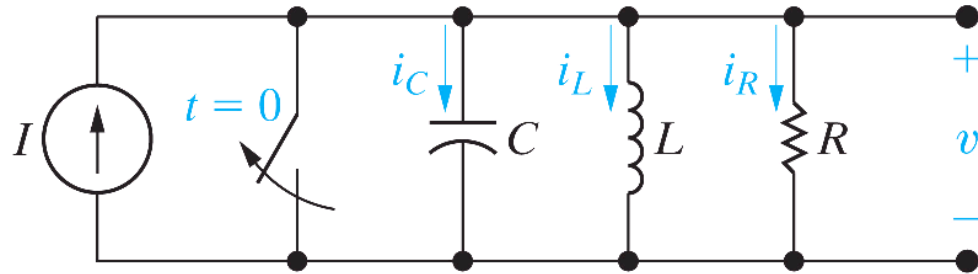
Make neat page or two of notes with equations for natural response and step response, for parallel RLC circuits.

Do the same for series RLC circuits.

Read all the examples

Do all the drill exercises.

Example: Find $v(t)$ for $t \geq 0$



Given

$$R = 250 \, (\Omega) \quad I_0 = 0.5 \, (A) = i_L(0)$$

$$L = 0.32 \, (H) \quad V_0 = 80 \, (V) = v_c(0)$$

$$C = 2 \, (\mu F) \quad I = -1.5 \, (A)$$

Current source switched in at $t = 0$

$$\alpha = \frac{1}{2RC} = \frac{1}{2(250)(2 \times 10^{-6})} = 1000 \, \frac{rad}{s} = \alpha$$

$$\omega_0 = \frac{1}{\sqrt{LC}} = \frac{1}{\sqrt{0.32(2 \times 10^{-6})}} = 1250 \, \frac{rad}{s} = \omega_0$$

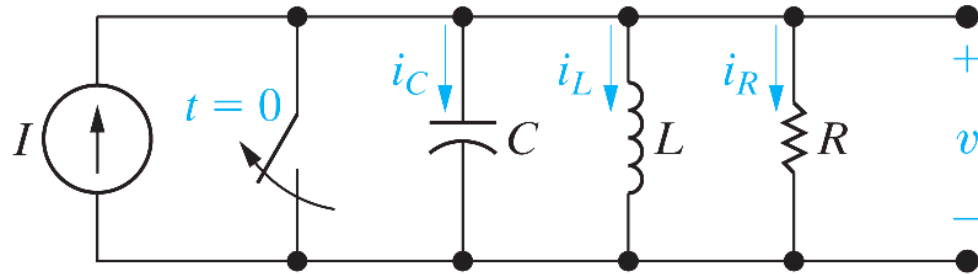
Find α and ω_0

$$\omega_0^2 > \alpha^2 \Rightarrow \text{Underdamped response}$$

$$\text{Note } i_L(\infty) = I = I_f$$

All steady state current is in inductor

Example (Contd.)



Solution is of the form

$$a) \quad i_L(t) = e^{-\alpha t} [B'_1 \cos(\omega_d t) + B'_2 \sin(\omega_d t)] + I_f$$

$$I_f = I = -1.5 \text{ (A)}$$

$$b) \quad s_{1,2} = -\alpha \pm j\omega_d$$

$$\omega_d = \sqrt{\omega_0^2 - \alpha^2} = \sqrt{1250^2 - 1000^2} = 750 \quad \left. \vphantom{\omega_d} \right\} \text{ Calculate the roots of the characteristic equation}$$

$$s_{1,2} = -1000 \pm j750 \frac{\text{rad}}{\text{sec}} \quad \left. \vphantom{s_{1,2}} \right\} \text{ Underdamped Solution}$$

$$c) \quad v_L = L \frac{di}{dt} \Rightarrow \frac{di_L(0^+)}{dt} = \frac{v_L(0^+)}{L} = \frac{V_0}{L} = \frac{80 \text{ (V)}}{0.32 \text{ (H)}} = 250 \left(\frac{\text{A}}{\text{s}} \right) \quad \left. \vphantom{\frac{di_L(0^+)}{dt}} \right\}$$

Given

$$R = 250 \text{ (}\Omega\text{)} \quad I_0 = 0.5 \text{ (A)} = i_L(0)$$

$$L = 0.32 \text{ (H)} \quad V_0 = 80 \text{ (V)} = v_c(0)$$

$$C = 2 \text{ (}\mu\text{F)} \quad I = -1.5 \text{ (A)}$$

Underdamped Solution

Find $\alpha, \omega_d, B'_1, B'_2$

Find Initial Conditions

$$i_L(0) = 0.5 \text{ (A)}$$

$$\frac{di_L(0^+)}{dt} = 250 \left(\frac{\text{A}}{\text{s}} \right)$$

Example (Contd.)

d) Solution has the form

$$i_L(t) = e^{-1000t} [B_1' \cos 750t + B_2' \sin 750t] - 1.5$$

$$i_L(0^+) = B_1' - 1.5 \Rightarrow i_L(0^+) = 0.5(A) \Rightarrow B_1' = 2(A)$$

e) Differentiate $i_L(t)$ and substitute $t = 0^+$ to find B_2'

$$\frac{di_L(t)}{dt} = -1000e^{-1000t} [B_1' \cos 750t + B_2' \sin 750t] + e^{-1000t} [-750B_1' \sin 750t + 750B_2' \cos 750t]$$

$$\frac{di_L(0^+)}{dt} = -1000B_1' + 750B_2' \quad \left\{ \begin{array}{l} \text{Use} \\ \frac{di_L(0^+)}{dt} = 250(A/s) \\ B_1' = 2(A) \end{array} \right. \Rightarrow B_2' = 3(A)$$

f) Solution has the final form

$$i_L(t) = e^{-1000t} [2 \cos 750t + 3 \sin 750t] - 1.5(A)$$

Example (Contd.)

g) $i_L(t) = e^{-1000t} [2 \cos(750t) + 3 \sin(750t)] - 1.5 \text{ (A)}$

Note: $i_L(0^+) = 2 - 1.5 = 0.5 \text{ A} \equiv I_0$ 

$i_L(\infty) = -1.5 \text{ (A)} = I_f \equiv I$ 

Perform as a
Check

h) We know $i_L(t)$ and $v(t)$ is the same for each element.

$\therefore v(t) = L \frac{di_L}{dt}$ } Differentiate $i_L(t)$ of (g) and multiply by L

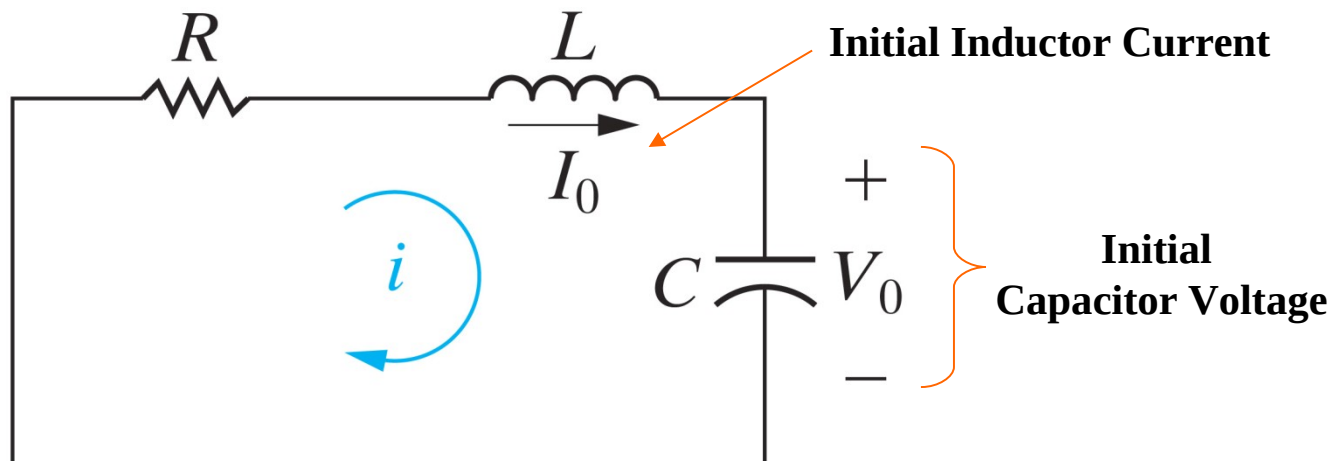
Final
Solution

$v(t) = 80e^{-1000t} [\cos(750t) - 18 \sin(750t)] \text{ (V)}$

After
Simplification

Note: $v(0) = 80 = V_0$ 

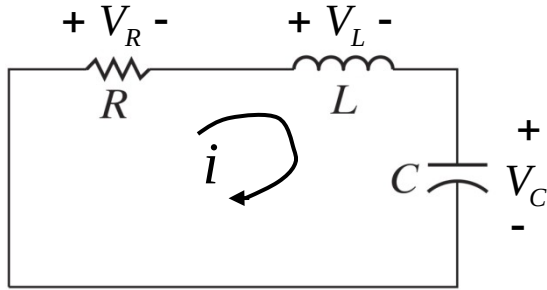
Series RLC Circuits



Similar to Parallel RLC Analysis

Find Natural Response for $i(t)$ by
obtaining Differential Equation in terms of $i(t)$.
Note that $i(t)$ is the same for all elements.

Series RLC Circuits (Contd.)



$$V_R + V_L + V_C = 0 \quad \left. \vphantom{V_R + V_L + V_C = 0} \right\} \text{KVL}$$

$$Ri + L \frac{di}{dt} + \left[\frac{1}{C} \int_{t_0}^t i(t) dt + V_0 \right] = 0 \quad \left. \vphantom{Ri + L \frac{di}{dt} + \left[\frac{1}{C} \int_{t_0}^t i(t) dt + V_0 \right] = 0} \right\} \begin{array}{l} \text{Substitute Ohm's Law} \\ \text{Capacitor Law} \\ \text{Inductor Law} \end{array}$$

Differentiate to get rid of integral

$$\Rightarrow \frac{d^2 i}{dt^2} + \frac{R}{L} \frac{di}{dt} + \frac{i}{LC} = 0$$

Characteristic Equation

$$s^2 + \left(\frac{R}{L} \right) s + \frac{1}{LC} = 0$$

Same form as for parallel RLC

Series RLC Circuits (Contd.)

$$i(t) = A_1 e^{s_1 t} + A_2 e^{s_2 t}$$

$$(\omega_0^2 < \alpha^2 \Rightarrow \text{overdamped})$$

$$i(t) = B_1 e^{-\alpha t} \cos(\omega_d t) + B_2 e^{-\alpha t} \sin(\omega_d t)$$

$$(\omega_0^2 < \alpha^2 \Rightarrow \text{underdamped})$$

$$i(t) = D_1 t e^{-\alpha t} + D_2 e^{-\alpha t}$$

$$(\omega_0^2 = \alpha^2 \Rightarrow \text{critically damped})$$

**Solution of
Differential
Equation**

$$\text{where: } s_{1,2} = -\alpha \pm \sqrt{\alpha^2 - \omega_0^2}$$

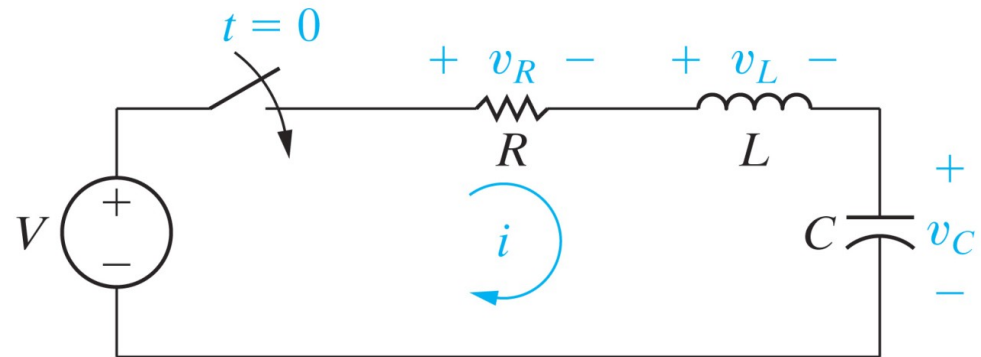
$$\alpha = R / 2L \quad \left. \vphantom{\alpha = R / 2L} \right\} \text{ Differs from } \parallel \text{ RLC}$$

$$\omega_0 = 1 / \sqrt{LC} \quad \left. \vphantom{\omega_0 = 1 / \sqrt{LC}} \right\} \text{ Same as } \parallel \text{ RLC}$$

$$\omega_d = \sqrt{\omega_0^2 - \alpha^2}$$

Series RLC Circuits: Step Response (Contd.)

Apply a DC Voltage Source
Focus on v_C since it is continuous



Similar to Parallel RLC analysis

$$v_R + v_L + v_C = V$$

} KVL

$$\textcircled{1} \quad Ri + L \frac{di}{dt} + v_C = V$$

} Substitute Ohm's Law and inductor law

Series RLC Circuits (Contd.)

Differentiate capacitor law

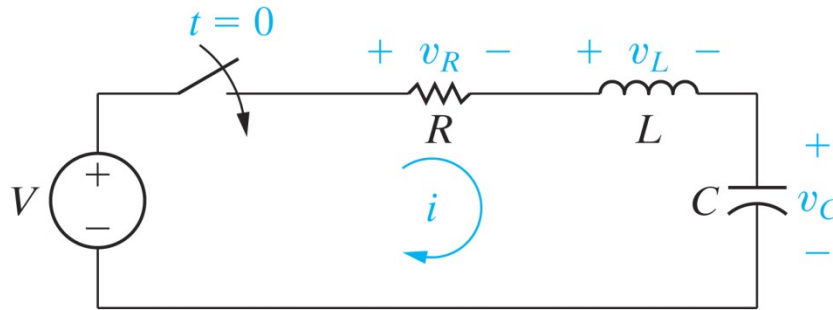
$$i = C \frac{dv_c}{dt} \Rightarrow \frac{di}{dt} = C \frac{d^2 v_c}{dt^2} \quad (2)$$

$$Ri + L \frac{di}{dt} + v_c = V \quad (1) \quad \left. \vphantom{\frac{di}{dt}} \right\} \text{From the previous slide}$$

$$\frac{d^2 v_c}{dt^2} + \frac{R}{L} \frac{dv_c}{dt} + \frac{v_c}{LC} = \frac{V}{LC} \quad \left. \vphantom{\frac{dv_c}{dt}} \right\} \text{Substitute } (2) \text{ into } (1)$$

Same form as for || RLC circuit

Series RLC Circuits: Step response for $v_C(t)$



$$V_f = V_{\text{cap}(\text{final})} = V$$

C - open; L - short

In steady state

$$v_C(t) = A'_1 e^{s_1 t} + A'_2 e^{s_2 t} + V_f$$

overdamped

Need

$s_1, s_2,$
 $\alpha, A'_1, A'_2,$

$B'_1, B'_2,$

D'_1, D'_2, ω_d

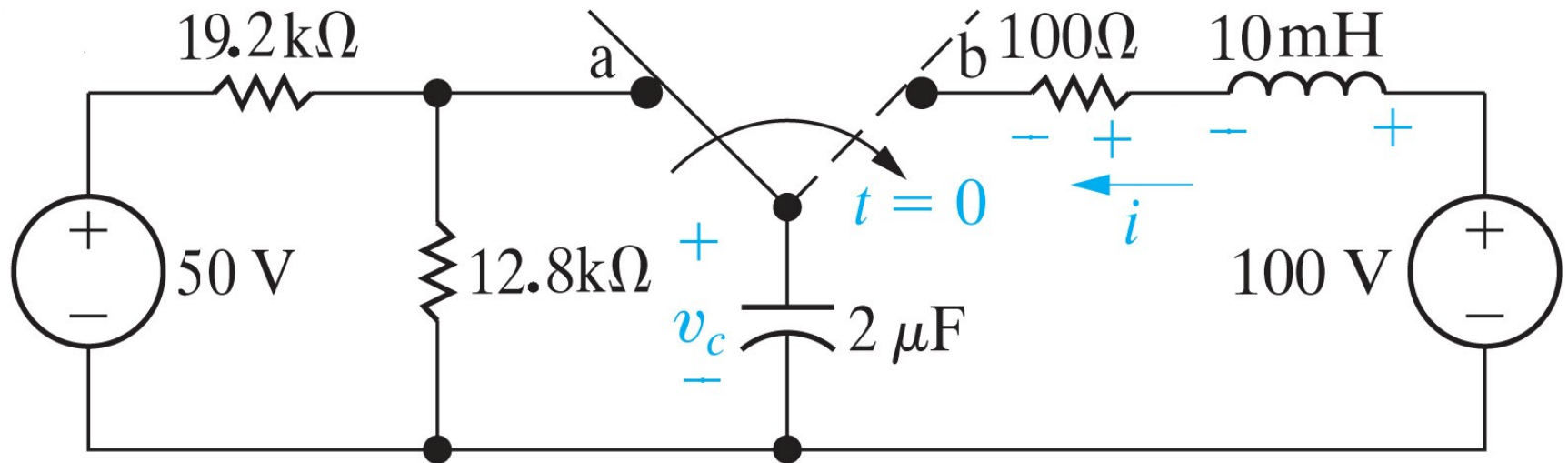
$$v_C(t) = e^{-\alpha t} [B'_1 \cos(\omega_d t) + B'_2 \sin(\omega_d t)] + V_f$$

underdamped

$$v_C(t) = D'_1 t e^{-\alpha t} + D'_2 e^{-\alpha t} + V_f$$

critically damped

Example



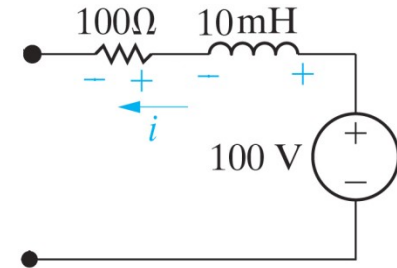
Find $i(t)$ for $t \geq 0$

Example: Find Initial Conditions (Contd.)

a) $i(0^-) = 0$

$i(0^-) = i(0^+) = 0$

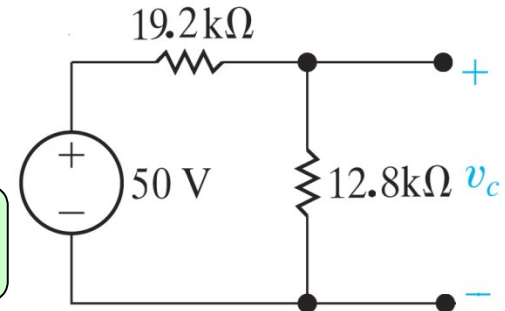
Since i_L can't change instantaneously



b) Since capacitor is open at $t = 0^-$

$$v_C(0^-) = \left(\frac{12.8K}{12.8K + 19.2K} \right) 50V = 20(V)$$

Voltage Division



$v_C(0^-) = v_C(0^+) = 20(V)$

Since v_c can't change instantaneously

Example: Find Parameters (Contd.)

c) $\alpha = \frac{R}{2L} = \frac{100}{2(10 \times 10^{-3})} = 5000 \text{ rad / s}$

$\omega_0 = \frac{1}{\sqrt{LC}} = \frac{1}{[(10 \times 10^{-3})(2 \times 10^{-6})]^{1/2}} = 7071.07 \text{ rad / s}$

Find α and ω_0

$$\omega_0 > \alpha^2 \Rightarrow \text{Underdamped}$$

$$\omega_d = \sqrt{\omega_0^2 - \alpha^2} = \sqrt{(7071.07)^2 - (5000)^2}$$

$\omega_d = 5000 \text{ rad / s}$ } Find ω_d

$$s_{1,2} = -\alpha \pm \sqrt{\alpha^2 - \omega_0^2} = -\alpha \pm j\omega_d$$

Find Characteristic equation roots

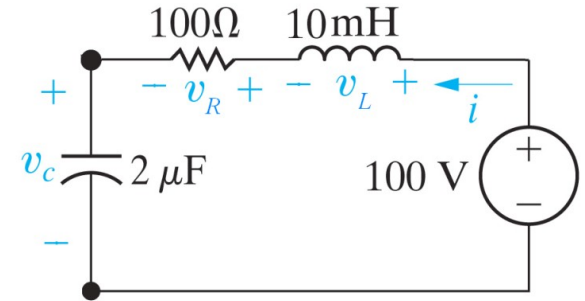
$$s_{1,2} = -5000 \pm j5000$$

Example: Find $i(t)$ (Contd.)

d)
$$i(t) = e^{-\alpha t} \left[B_1' \cos(\omega_d t) + B_2' \sin(\omega_d t) \right] + I_f \quad \left. \vphantom{i(t)} \right\} \text{General solution for Underdamped system}$$

$I_f = 0$ since eventually capacitor becomes open

e) We know α , ω_d , and I_f
We must use $i(0^+)$ and $\frac{di(0^+)}{dt}$ to find B_1' and B_2'

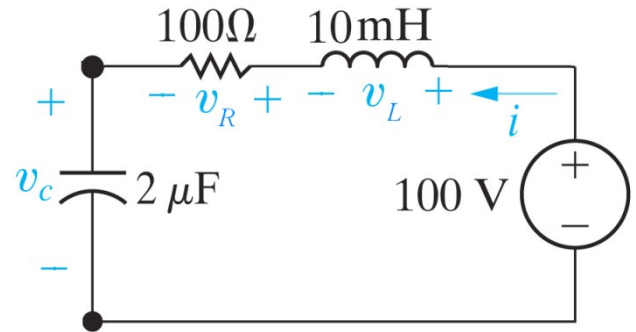


f) Find $i(0^+)$
 $i(0^+) = 0 \quad \left. \vphantom{i(0^+)} \right\} \text{From a previous slide}$

Example: Find Initial Conditions (Contd.)

g) Find $\frac{di(0^+)}{dt}$

Circuit at $t = 0^+$



Need $\frac{di(0^+)}{dt} \Rightarrow v_L(0^+) = L \frac{di(0^+)}{dt}$ USE

From a previous slide

KVL $\{ 100V = v_L + v_R + v_C$

$i(0^+) = 0 \Rightarrow v_R(0^+) = 0$

Ohm's Law

$100 = v_L(0^+) + v_C(0^+)$

$v_C(0^+) = 20$
from the previous slide

$100 = L \frac{di(0^+)}{dt} + 20$

Substitute inductor law

$\frac{di(0^+)}{dt} = \frac{100 - 20}{L} = \frac{80}{10 \times 10^{-3}}$

Solve for $\frac{di(0^+)}{dt}$

$\frac{di(0^+)}{dt} = 8000 \text{ (A/s)}$

Example: Find $i(t)$ (Contd.)

h)

Find B_1' and B_2' in general solution

$$i(t) = e^{-\alpha t} [B_1' \cos(\omega_d t) + B_2' \sin(\omega_d t)] + I_f$$

General Solution

$$i(0^+) = 0 \quad \left\{ \begin{array}{l} \text{From a previous slide} \end{array} \right.$$

$$0 = B_1' + I_f \quad \left\{ \begin{array}{l} \text{Substitute } t = 0^+ \end{array} \right.$$

$$B_1' = 0 \quad \left\{ \begin{array}{l} \text{Since } I_f = 0 \end{array} \right.$$

Take time derivative

$$\frac{di(t)}{dt} = -\alpha e^{-\alpha t} [B_1' \cos(\omega_d t) + B_2' \sin(\omega_d t)] + e^{-\alpha t} [-B_1' \omega_d \sin(\omega_d t) + B_2' \omega_d \cos(\omega_d t)]$$

$$\frac{di(0^+)}{dt} = -\alpha B_1' + \omega_d B_2' = 8000 \quad \left\{ \begin{array}{l} \text{Substitute } t = 0^+ \text{ and use value for } \frac{di(0^+)}{dt} \end{array} \right.$$

$\square \square \square \square \square \square \square \square \square \square$

$$B_1' = 0, \alpha = 500, \omega_d = 5000 \Rightarrow B_2' = 1.6(A)$$

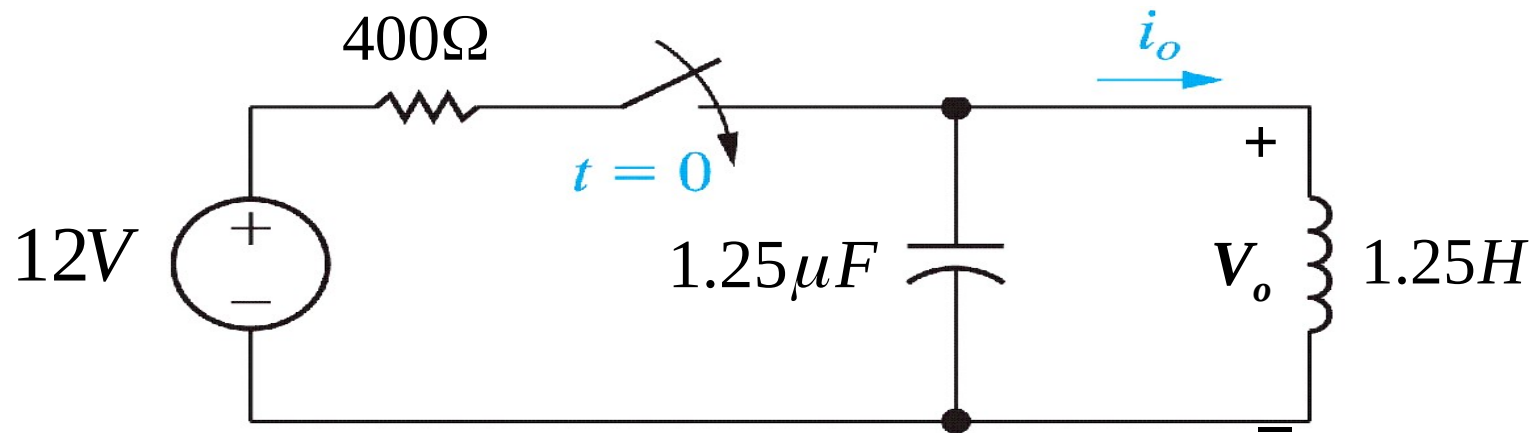
Final Solution

$$i(t) = 1.6e^{-5000t} \sin(5000t)(A) \quad t \geq 0$$

Check

$$i(0) = 0, i(\infty) = 0$$

Example: Find $V_o(t)$ for $t \geq 0$



No initial energy stored in the circuit

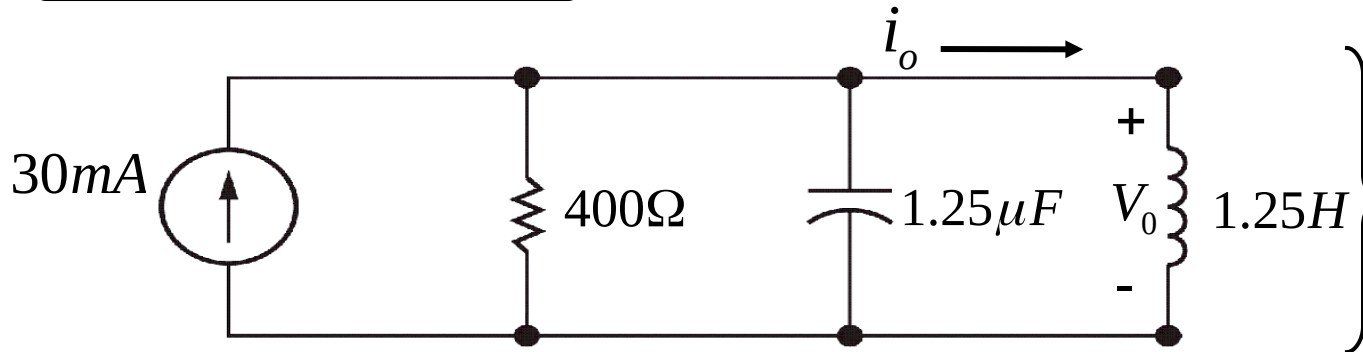
Given in the problem

$$\begin{aligned} i_o(0^-) &= i_o(0^+) = 0 \\ V_o(0^-) &= V_o(0^+) = 0 \end{aligned}$$

Properties of
inductors and
capacitors

Example: Use Source Transformation (Contd.)

Redraw the circuit for $t \geq 0$



Source Transformation

$$30(mA) = \frac{12(V)}{400(\Omega)}$$

Parallel *RLC*

$$\alpha = \frac{1}{2RC} = \frac{1}{2(400)(1.25 \times 10^{-6})} = 1000$$

$$\omega_0 = \frac{1}{\sqrt{LC}} = \frac{1}{\sqrt{1.25(1.25 \times 10^{-6})}} = \sqrt{64 \times 10^4} = 800$$

Find α and ω_0

$\alpha > \omega_0$ overdamped

Example (Contd.)

$$\left. \begin{aligned} s_{1,2} &= -\alpha \pm \sqrt{\alpha^2 - \omega_0^2} = -1000 \pm 600 \\ s_1 &= -400, \quad s_2 = -1600 \end{aligned} \right\} \begin{array}{l} \text{Calculate the roots of the} \\ \text{Characteristic equation} \end{array}$$

$$V_o(t) = V_f + A_1 e^{s_1 t} + A_2 e^{s_2 t} \quad \left\{ \begin{array}{l} \text{Overdamped solution has this form} \end{array} \right.$$

$$V_f = 0 \quad \left\{ \begin{array}{l} \text{C open, L short, all current in L} \end{array} \right. \quad \left\{ \begin{array}{l} V_o = 0 \text{ as } t \rightarrow \infty \end{array} \right.$$

$$V_o(t) = A_1 e^{-400t} + A_2 e^{-1600t} \quad \left\{ \begin{array}{l} \text{Use initial conditions to find } A_1 \text{ and } A_2 \end{array} \right.$$

Now must find $V_o(0^+)$ and $\frac{dV_o(0^+)}{dt}$ to solve for A_1 and A_2

Example (Contd.)

Find $V_o(0^+) \Rightarrow V_o(0^+) = 0$ } Given in the problem

Find $\frac{dV_o(0^+)}{dt}$ using the circuit

$V_o(0) = V_o(0^+) = 0$ } Voltage across capacitor can't change instantly

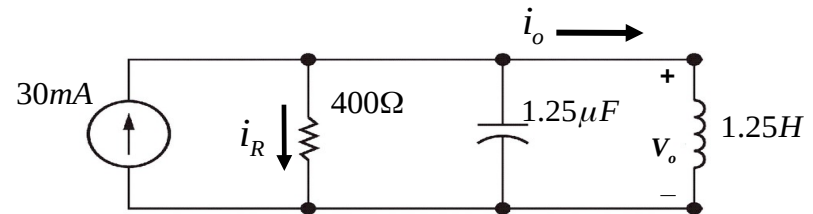
$i_R(0^+) = V_o/R = 0$ } Ohm's Law

$30mA = i_R(0^+) + i_C(0^+) + i_o(0^+)$ } KCL

$30mA = i_C(0^+)$ } since $i_R(0^+) = i_o(0^+) = 0$

$\frac{dV_o(0^+)}{dt} = \frac{1}{C} i_C$ } Capacitor Law

$\frac{dV_o(0^+)}{dt} = \frac{i_C(0^+)}{C} = \frac{30mA}{1.25\mu F} = 24,000V/s$ } Plug in numbers



Example (Contd.)

Find A_1 and A_2

$$V_0(t) = A_1 e^{-400t} + A_2 e^{-1600t} \quad \left. \vphantom{V_0(t)} \right\} \text{Solution so far}$$

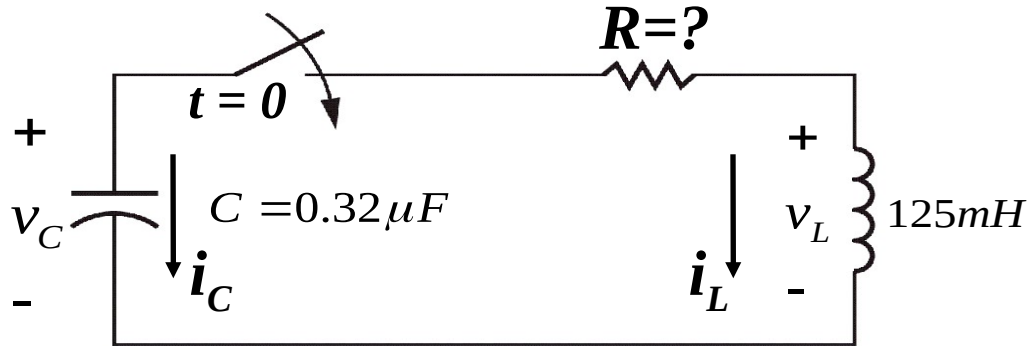
$$\textcircled{1} \quad V_0(0^+) = 0 = A_1 + A_2 \quad \left. \vphantom{V_0(0^+)} \right\} \text{Plug in } t = 0^+$$

$$\textcircled{2} \quad \frac{dV_0(0^+)}{dt} = 24,000 = -400A_1 - 1600A_2 \quad \left. \vphantom{\frac{dV_0(0^+)}{dt}} \right\} \begin{array}{l} \text{Differentiate } v_0(t) \text{ and} \\ \text{substitute } t = 0^+ \end{array}$$
$$\Rightarrow A_1 = 20V \quad A_2 = -20V \quad \left. \vphantom{\Rightarrow} \right\} \begin{array}{l} \text{Solve } \textcircled{1} \text{ and } \textcircled{2} \\ \text{for 2 unknowns} \end{array}$$

**Final
Solution**

$$V_0(t) = 20 \left[e^{-400t} - e^{-1600t} \right] (V)$$

Example: Design R so that $v_c(t)$ is critically damped



Given

$$v_C(0^+) = 15(V)$$

$$i_L(0^+) = 6(mA)$$

a) Critical Damping: $\alpha = \omega_0$

$$\text{Series} \rightarrow \frac{R}{2L} = \frac{1}{\sqrt{LC}} \Rightarrow R = 1250(\Omega)$$

b) $i_L(0^+) = i_L(0^-) = 6(mA)$ } Current cannot change instantly

Series RLC Current same in all components

Example (Contd.)

c) $\alpha = \frac{R}{2L} = \frac{1250}{2(0.125)} = 5000 \frac{\text{rad}}{\text{s}}$ } Calculate α

d) $v_c(t) = D_1 t e^{-5000t} + D_2 e^{-5000t}$ } General expression for critically damped system

e) We must find $v_c(0^+)$ and $\frac{dv_c(0^+)}{dt}$ to solve for D_1 and D_2

f) $v_c(0^+) = 15(\text{V})$ } Given in the problem

g) $i_c(t) = C \frac{dv_c(t)}{dt}$ } Capacitor Law

so $\frac{dv_c(0^+)}{dt} = \frac{1}{C} i_c(0^+) = \frac{-i_L(0^+)}{32(\mu\text{F})}$ } From Circuit
 $i_c = -i_L$

so $\frac{dv_c(0^+)}{dt} = -18,750(\text{A/F})$ } Substitute $i_L(0^+) = 6(\text{mA})$

Example (Contd.)

h) Find D_1 and D_2

$$v_C(t) = D_1 t e^{-5000t} + D_2 e^{-5000t} \quad \left. \vphantom{v_C(t)} \right\} \text{General Solution}$$

$$v_C(0^+) = 15(V) \quad \frac{dv_C(0^+)}{dt} = -18,750(A/F) \quad \left. \vphantom{v_C(0^+)} \right\} \text{From previous slide}$$

$$v_C(0^+) = 15(V) = D_2 \quad \left. \vphantom{v_C(0^+)} \right\} \text{Substitute } t = 0^+$$

$$\frac{dv_C(t)}{dt} = D_1 e^{-5000t} - 5000t D_1 e^{-5000t} - 5000 D_2 e^{-5000t} \quad \left. \vphantom{\frac{dv_C(t)}{dt}} \right\} \text{Time derivative of } v_C(t)$$

$$\frac{dv_C(0^+)}{dt} = -18,750 = D_1 - 5000 D_2 \quad \left. \vphantom{\frac{dv_C(0^+)}{dt}} \right\} \text{Substitute } t = 0^+$$

$$D_1 = 56,250(V/s) \quad \left. \vphantom{D_1} \right\} \text{Substitute } D_2 = 15$$

$$v_C(t) = 56,250 e^{-5000t} + 15 e^{-5000t} \quad \left. \vphantom{v_C(t)} \right\} \text{Final Solution}$$